XIAMEN UNIVERSITY MALAYSIA

FINAL EXAMINATION

Course Code:	COM107		
Course Name:	Circuits and Devices		
Question Paper Setter:	Chua Ming Yam		
Academic Session:	2022/04	Question Paper:	A ☑ B ☐
Total No. of Pages:	4	Time Allocated:	2 hours
Additional Materials:	All materials related to COM107		
Apparatus Allowed:	-		

INSTRUCTIONS TO CANDIDATES

1. This paper consists of **FOUR** pages excluding the cover page.
2. There are **FOUR** questions in this paper. Attempt **ALL**.
3. Please follow the requirement of each question and type in all the corresponding answers on the downloadable answer book provided.
4. Communication between candidates in any means is forbidden. Answers must be entirely individual candidate's independent effort. If you are found sharing your solutions with other candidates, or suspected of doing so, you would be penalized accordingly.

(Student ID: _____ Full Name: _____)

Question 1 (25%)

A circuit shown in Figure Q1 is supplied with an a.c. source, $V1$, with an instantaneous voltage of,

$$v_1(t) = 240\cos(314.16t - 90^\circ)$$

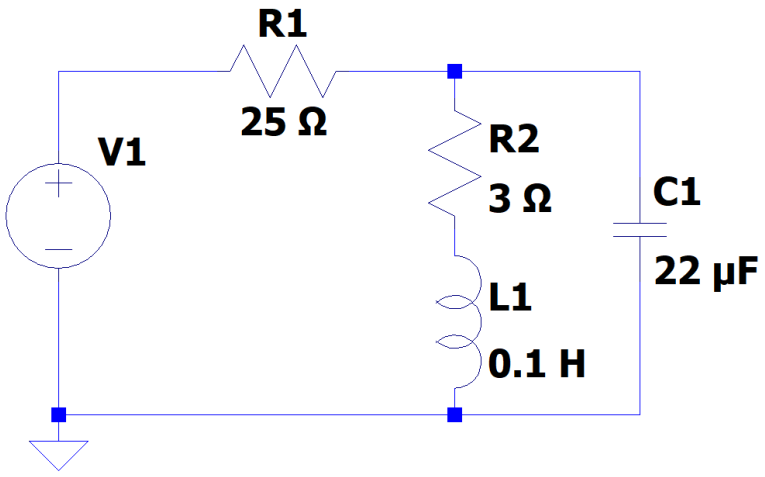


Figure Q1

a) Derive the total impedance, Z_T, of the circuit. Express it in complex form. [7 marks]

b) Determine the source current I. Express it in complex form. [6 marks]

c) Determine the phase angle of the circuit. [4 marks]

d) Tune the phase angle of the circuit to 60°. [8 marks]

Continues….

Question 2 (25%)

A machine with a coil having an internal resistance of 3 Ω and an inductance of 0.1 H is supplied by an a.c. source of 240 V, 50 Hz. Figure Q2 shows the equivalent circuit of the machine.

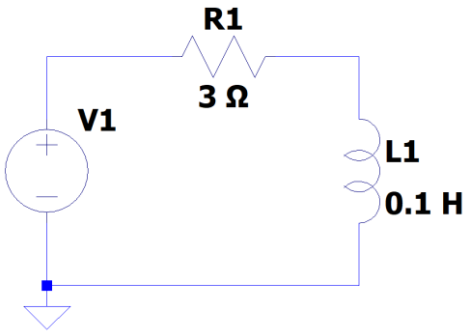


Figure Q2

a) Evaluate the power rating requirement of the machine. [6 marks]

[Hint: Power rating is the total amount of power required by a machine to work under normal situation.]

b) Evaluate the energy efficiency of the machine. [6 marks]

[Hint: Power factor is an expression of energy efficiency.]

c) Verify your evaluation in Q2(b) by determining the active and reactive power in the circuit. [6 marks]

d) Design a power factor correction circuitry to improve the machine efficiency to at least 90%. [7 marks]

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Question 3 (25%)

A tuned LC circuits shown in Figure Q3 has an inductor of 1.0 nH and a variable capacitor with the capacitance in the range of 1 nF to 20 nF.

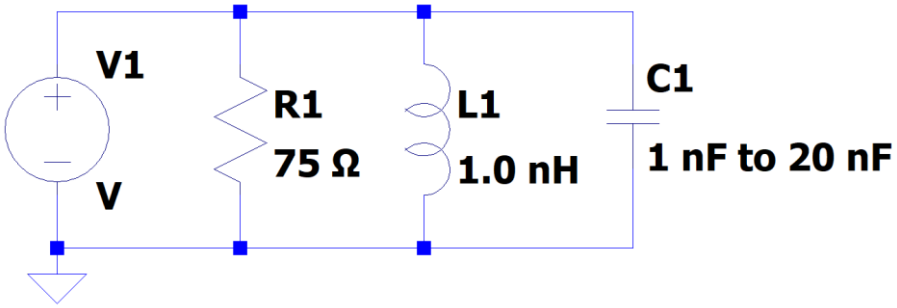


Figure Q3

a) Tune the circuit to resonant at 98.8 MHz. Explain how the circuit behave at resonance. [11 marks]

b) Determine the Q factor of the tuned circuit. What does it mean? [6 marks]

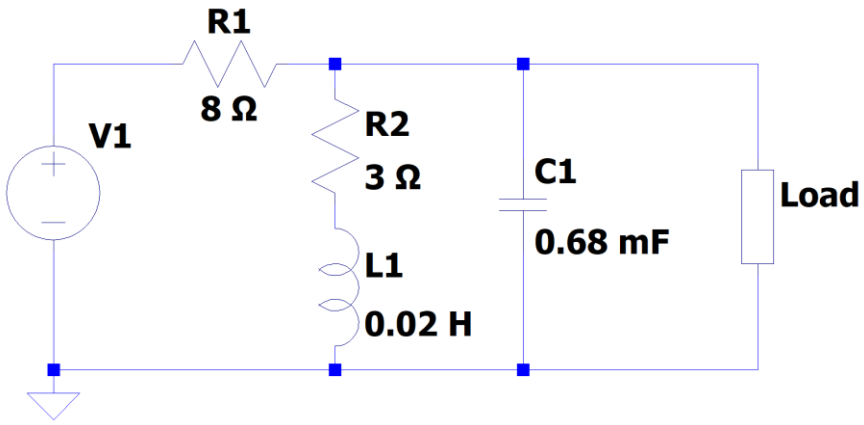
c) How much is the bandwidth of the circuit? [3 marks]

d) What are the half-power frequencies? [5 marks]

Continues….

Question 4 (25%)

a) For the circuit given in Figure Q4a, the source voltage is $240V$, $50\ Hz$.



Convert the circuit to its Thevenin equivalent circuit. [15 marks]

b) Design a bandpass filter with a lower cutoff frequency of $9\ MHz$ and upper cutoff frequency of $11\ MHz$. Show all your calculations. [10 marks]

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